INCH-POUND

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SUPERSEDING
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AMENDMENT 4
03 January 1983

DETAIL SPECIFICATION

ACRYLIC CLOTH, IGNITER

Reactivated after 31 July 2026 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the U.S. Army Armaments Center and is available for use by all Departments and Agencies of the Department of War within the distribution limitations noted at the bottom of the page.

1. SCOPE

1.1. Scope. This specification covers one type of igniter cloth for use in the manufacture of igniter pads and bags in ammunition. The igniter cloth is acrylic (breaking strength 100lbs/ in minimum).

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM AC ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1395

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2. APPLICABLE PUBLICATIONS

2.1. General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as example. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of piblications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government publications.

2.2.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Department of War Standards

MIL-STD-1916 – DoD Preferred Methods for Acceptance of Product
(Copies of this publication are available online https://assist.quicksearch.dla.mil)

2.3. Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation of the contract.

ASTM International

ASTM D1776	–	Standard Practice for Conditioning Textiles
ASTM D1777	–	Standard Test Method for Thickness of Textile Materials
ASTM D2257	–	Standard Test Method for Extractable Matter in Textiles
ASTM D2594/D2594M	–	Standard Test Method for Stretch Properties of Knitted Fabrics Having Low Power
ASTM D3775	–	Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776	–	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D1683	–	Standard Test Method for Failure in Sewn Seams Woven Fabric
ASTM D5034	–	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D5630	–	Standard Test Method for Ash Content in Plastics
ASTM D7309	–	Standard Test Method for Determining Flammability Characteristics of Plastics and

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Other Solid Materials Using Microscale Combustion Calorimetry

(Copies of these publications are available online at https://www.astm.org)

American Society of Mechanical Engineers (ASME)

SAE-AMS-STD-595 – Colors Used in Government Procurement

(Copies of this publication are available online at https://www.sae.org)

International Organization for Standardization (ISO)

ISO 3071 – Determination of pH of Aqueous Extract

(Copies of this publication are available online at https://www.iso.org)

2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. Conformance inspection. Conformance inspection (also known as Lot Acceptance Testing (LAT)) shall be performed for every lot of material in accordance with 4.2.

3.2. Material type. The igniter cloth shall be woven from acrylic yarns spun from acrylic fibers containing no less than 100% polyacrylonitrile.

3.3. Absence of mono-polymer. No mono-polymer shall be used.

3.4. Width. The width of the cloth and its tolerance shall be as specified in the contract or purchase order (see 6.2).

3.5. Thickness. The thickness of the cloth and its tolerance shall be as specified in the contract or purchase order (see 6.2).

3.6. Weave. The cloth, either woven one or two in a dent, as specified in the contract or purchase order (see 6.2) shall be a plain single weave having the same number of ends and picks per inch. The warp and filling yarns shall be of the same size.

3.7. Slippage. The slippage value shall be a value not less than 60 for warp and fill.

3.8. Stretch. The percent stretch shall be a value not greater than 10.0 for warp and fill.

3.9. Ash. When charred, the ignitor cloth ash shall be a value not greater than 2.0 percent per original weight.

3.10. Acidity. The acidity shall be as follows: acidity is determined to be a value not greater than acetic acid 0.1 percent.

3.11. Alkalinity. The alkalinity shall be as follows: alkalinity is determined to be a value not greater than sodium carbonate 0.1 percent.

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3.12. Extractable material. The ether soluble material shall be a value not greater than 5 percent.

3.13. Halogens. No halogens shall be present in the material.

3.14. Flammability characteristics. The character and quality of the material shall have no indication of smoldering when the cloth is ignited sufficiently to give a good flame and then extinguished.

3.15. Color. The cloth shall either be dyed to be closely equivalent to either red color number 11105 or natural white color number 27880 as per SAE-AMS-STD-595 as specified in the contract or purchase order (see 6.2).

3.16. Cloth weight. The weight of the cloth shall be a value not less than 5.25 ounces (oz.) per square yard (sq. yd.) or not greater than 5.75 oz. per sq yd.

3.17. Breaking strength. The breaking strength of both the warp and fill of the cloth shall be a value not less than 100 pounds (lb.) per inch.

3.18. Thread count. The thread count shall neither be a value not less than 35 threads per in nor a value not greater than 45 threads per inch.

3.19. Workmanship. The cloth shall be uniformly and closely woven and shall be free from defects.

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4. VERIFICATION

TABLE 1. Requirement/ verification cross reference matrix.

Method of Verification 1- Analysis 2- Demonstration 3- Examination 4- Test		Classes of verification A- Conformance inspection					
Section 3 Requirement		Verification Methods				Verification Class	Section 4 Verification Procedures
Paragraph number	Title	1	2	3	4	A	
3.1	LAT	X	X	X	X	X	4.2
3.2	Material type	X		X		X	4.4
3.3	Absence of mono-polyme	X		X		X	4.5
3.4	Width	X				X	4.6
3.5	Thickness	X				X	4.7
3.6	Weave	X				X	4.4 & 4.18
3.7	Slippage	X				X	4.8
3.8	Stretch	X				X	4.9
3.9	Ash	X				X	4.10
3.10	Acidity	X				X	4.11
3.11	Alkalinity	X				X	4.12
3.12	Extractable material	X				X	4.13
3.13	Halogens			X		X	4.14
3.14	Flammability characteristi			X		X	4.14
3.15	Color			X		X	4.15
3.16	Cloth weight	X				X	4.16
3.17	Breaking strength	X				X	4.17
3.18	Thread count	X				X	4.18
3.19	Workmanship			X		X	4.19

4.1. Classification of verification. The verification requirements specified herein are classified as follows:

a. LAT (see 4.2)

4.2. LAT. LAT shall be performed as specified in TABLE 1. All characteristics are classified as major characteristic except workmanship which is a minor characteristic.

4.2.1. Rejection. If any sample fails to meet the requirements, the lot shall be rejected.

4.2.2. Sampling. The sample unit shall be sufficient in length such that all physical and chemical test can be completed. The amount of sample units shall be based on the lot size. There shall be 2 sample units for lot sizes not greater than 800 yards, 3 sample units for lot size not less than 801 yards and not greater than 22,000 yard, and 5 sample units for lot sizes not less than 22,001 yards.

4.3. Test methods. The following paragraphs serve as references to help identify the quantities and tests needed to complete LAT. All test samples shall be conditioned in accordance with ASTM D1776.

4.4. Material type. The material type shall either be verified through a certificate of conformance from the original supplier or as demonstrated by an analytical test method.

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4.5. Absence of mono-polymer. The absence of mono-polymer shall either be verified through a certificate of conformance from the original supplier or as demonstrated by an analytical test method.

4.6. Width. The width of the fabric shall be taken as a part of any method from ASTM D3776.

4.7. Thickness. The thickness of the fabric shall be taken in accordance with ASTM D1777.

4.8. Slippage. The slippage values shall be taken in accordance with ASTM D1683. The reported value shall be the specific force at which yarn slippage or displacement occurs, to evaluate compliance with the slippage requirements specified in 3.7.

4.9. Stretch. The stretch value shall be taken in accordance with ASTM D2594/D2594M.

4.10. Ash. The ash content shall be taken in accordance with ASTM D5630.

4.11. Acidity. The acidity shall be taken in accordance with ISO 3071.

4.12. Alkalinity. The alkalinity shall be taken in accordance with ISO 3071.

4.13. Extractable material. The amount of extractable material shall be determined through a method that complies with ASTM D2257.

4.14. Flammability characteristics. A method that follows either method A or B from ASTM D7309 which records and reports information from each step and final report information in sections 12 and 13 of ASTM D7309.

4.15. Color. Color shall either be matched to color chips or, when in doubt, from color coordinates obtained from spectrophotometric analysis. Color chips shall not be any older than 3 years to account for fading.

4.16. Cloth weight. Cloth weight shall be determined by a method from ASTM D3776/D3776M and reported in oz. per sq. yd.

4.17. Breaking strength. Breaking strength shall be determined by a method from ASTM D5034 and reported in lb. per inch.

4.18. Thread count. Thread count shall be determined by a method from ASTM D3775.

4.19. Workmanship. Workmanship shall be completed through an inspection which shall ensure the fabric has no defect which could affect the use of the material when it is being processed into the igniter assembly.

5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or War Agency, or within the Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

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6.1. Intended use. The cloth covered by this specification is intended for use with artillery propelling charge igniter bags and pads. The cloth and its requirements covered by this specification are military unique since artillery propelling charges have no commercial application.

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Requirements for submission of first article sample
c. Requirements for submission of lot acceptance sample
d. Certificates of conformance for all test methods mentioned within this specification
e. Contract inspection equipment designs for approval
f. Requirement for lot and interfix numbering
g. Certificate of compliance for individual constituent materials
h. Width of the cloth and its tolerance
i. Thickness of the cloth and its tolerance
j. Type of weave
k. Color of cloth
l. Sampling plan, if additional samples are requested

6.3. Submission of contract inspection equipment designs for approval. Submit copies of designs as required to: U.S. Army DEVCOM AC, ATTN: FCDD-ACE-QMA, Picatinny, NJ 07806-5000.

6.4. Inspection lot size. It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests for deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5. Lot formation. Lot formation should be IAW the requirements of MIL-STD-1916 or as otherwise specified in the contract or purchase order.

6.6. Submission of alternative conformance provisions. All contractor proposed alternative conformance provisions should be submitted to the Government for evaluation/ approval as directed by the contracting activity.

6.7. Equivalent test methods. The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to DEVCOM AC, ATTN: FCDD-ACE-QMA, Picatinny, NJ 07806-5000.

6.8. Subject term (key word) listing.

Fabric
Woven
Yarn

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6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Army-AR

Preparing activity:
Army-AR
(Project 1395-2022-001)

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